CTU: Count up

Description

You can use the "Count up" instruction to increment the value at output CV. When the signal state at the CU input changes from "0" to "1" (positive signal edge), the instruction is executed and the current counter value at the CV output is incremented by one. The counter value is incremented each time a positive signal edge is detected, until it reaches the high limit for the data type specified at the output CV. When the high limit is reached, the signal state at the CU input no longer has an effect on the instruction.

You can query the counter status in the Q output. The signal state at the Q output is determined by the PV parameter. If the current counter value is greater than or equal to the value of the PV parameter, the Q output is set to signal state "1". In all other cases, the Q output has signal state "0".

The value at the CV output is reset to zero when the signal state at input R changes to "1". As long as the R input has signal state "1", the signal state at the CU input has no effect on the instruction.

> **Note**
>
> Only use a counter at a single point in the program to avoid the risk of counting errors.

Each call of the "Count up" instruction must be assigned an IEC counter in which the instruction data is stored. An IEC counter is a structure with one of the following data types:

For S7-1200 CPU

Data block of system data type IEC_<Counter> (Shared DB)	Local tag
• IEC_SCOUNTER / IEC_USCOUNTER • IEC_COUNTER / IEC_UCOUNTER • IEC_DCOUNTER / IEC_UDCOUNTER	• CTU_SINT / CTU_USINT • CTU_INT / CTU_UINT • CTU_DINT / CTU_UDINT • IEC_SCOUNTER / IEC_USCOUNTER • IEC_COUNTER / IEC_UCOUNTER • IEC_DCOUNTER / IEC_UDCOUNTER

For S7-1500 CPU

Data block of system data type IEC_<Counter> (Shared DB)	Local tag
• IEC_SCOUNTER / IEC_USCOUNTER • IEC_COUNTER / IEC_UCOUNTER • IEC_DCOUNTER / IEC_UDCOUNTER • IEC_LCOUNTER / IEC_ULCOUNTER	• CTU_SINT / CTU_USINT • CTU_INT / CTU_UINT • CTU_DINT / CTU_UDINT • CTU_LINT / CTU_ULINT • IEC_SCOUNTER / IEC_USCOUNTER • IEC_COUNTER / IEC_UCOUNTER • IEC_DCOUNTER / IEC_UDCOUNTER • IEC_LCOUNTER / IEC_ULCOUNTER

You can declare an IEC counter as follows:

- Declaration of a data block of system data type IEC_<Counter> (for example, "MyIEC_COUNTER")
- Declaration as a local tag of the type CTU_<Data type> or IEC_<Counter> in the "Static" section of a block (for example #MyIEC_COUNTER)

When you set up the IEC counter in a separate data block (single instance), the instance data block is created by default with "optimized block access" and the individual tags are defined as retentive. For additional information on setting retentivity in an instance data block, refer to "See also".

When you set up the IEC counter as local tag (multi-instance) in a function block with "optimized block access", it is defined as retentive in the block interface.

The execution of the "Count up" instruction requires a preceding logic operation. It can be placed within or at the end of the network.

Parameters

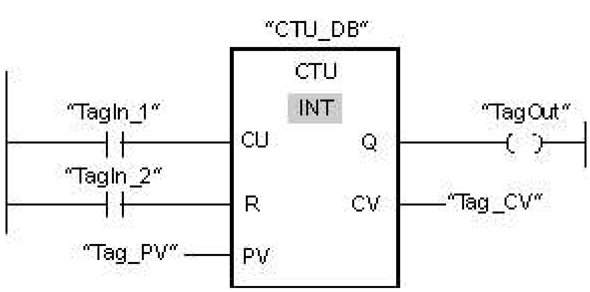
The following table shows the parameters of the "Count up" instruction:

Parameter	Declaration	Data type	Memory area		Description
			S7-1200	S7-1500	
CU	Input	BOOL	I, Q, M, D, L or constant	I, Q, M, D, L or constant	Count input
R	Input	BOOL	I, Q, M, D, L, P or constant	I, Q, M, T, C, D, L, P or constant	Reset input
PV	Input	Integers	I, Q, M, D, L, P or constant	I, Q, M, D, L, P or constant	Value at which the output Q is set.
Q	Output	BOOL	I, Q, M, D, L	I, Q, M, D, L	Counter status
CV	Output	Integers, CHAR, WCHAR, DATE	I, Q, M, D, L, P	I, Q, M, D, L, P	Current counter value

You can select the data type of the instruction from the "???" drop-down list of the instruction box.

Example

The following example shows how the instruction works:



When the signal state of the "TagIn_1" operand changes from "0" to "1", the "Count up" instruction is executed and the current counter value of the "Tag_CV" operand is incremented by one. With each additional positive signal edge, the counter value is incremented until the high limit of the data type (INT = 32767) is reached.

The value of the PV parameter is adopted as the limit for determining the "TagOut" output. The "TagOut" output has signal state "1" as long as the current counter value is greater than or equal to the value of the "Tag_PV" operand. In all other cases, the "TagOut" output has signal state "0".

For more information and the program code to the above named example, refer to: Sample Library for Instructions

---

See also

Overview of the valid data types
Setting the retentivity of local tags
Setting retentivity in an instance data block
Memory areas (S7-1500)
Basic information on LAD
Memory areas (S7-1200)